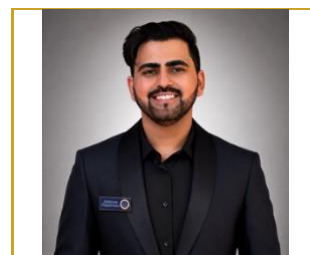


Dr. Vishal Chaudhary (He/Him/His)

Research Leader in Complex Life–Matter Interfacial Dynamics |
Sensor Intelligence | Nanomaterials for One Health

PHYSICIST • EDUCATOR • EDITOR • DATA & MATERIAL SCIENTIST
•MENTAL HEALTH ADVOCATE



Executive Profile

His research is centered on understanding, engineering, and decoding complex interactions at material, biological, environmental and societal interfaces. Beginning with nanoengineered materials and sensing platforms, his work has progressively expanded toward complex systems, data-driven modelling, and intelligent decision frameworks, culminating in the concept of Sensor Intelligence. This interdisciplinary vision combines advanced nanomaterials, sensing technologies, network science, complex physics, and machine intelligence to transform raw signals into actionable insights for healthcare, environmental sustainability, and One Health management.

Current Appointments

Lecturer, Physics

NAS, Mahidol University, Thailand

Leader

Sensor Intelligence Lab, NAS
Sensor Intelligence Network

Editor

Scientific Reports, Nature

Adjunct Professor

Chitkara University, India

Research Metrics Dashboard

(Google Scholar: Accessed 23 June 2026)

48 H-index

6889 Citations

175+ Publications

75.2% Main Author

Awards & Recognitions

- 🏆 World Top 2% Scientists (Stanford Ranking, 2023–2025)
- 🏆 Top 5% Scientist Worldwide (SciRank, 2025)
- 🏆 Agents of Change Awardee (Elsevier Foundation 2023)
- 🏆 SDG Service Award (ACS, ECS & IOP, 2021)
- 🏆 Silver Award, Country Coordinator (GSFN, UK)
- 🏆 Manav Sanrakshak Samman (University of Delhi, 2025)
- 🏆 Mahidol Best Research Paper Award 2025
- 🏆 All India Rank 171 in National Eligibility Test for Lectureship in 2012

Professional Information

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Contact:
+66 0831392120, +91 9971757034

Profile links:

[Google Scholar](#), [ORCID](#),
[Linkedin](#), [Scopus](#),
[Web of Science](#), [Loop](#), [SciProfiles](#)
[MureX](#), [NAS](#), [Research Gate](#)

Education

Ph. D., Department of Physics & Astrophysics

University of Delhi, New Delhi, India

Thesis title: *Study of Polyaniline & its Nanocomposites for Gas Sensing Applications*

10/2012 – 06/2019

M.Sc. Physics, Kirori Mal College

University of Delhi, New Delhi, India

Area of Specialization: Physics at the Nanoscale

07/2010 – 06/2012

B.Sc. (Honours) Physics, Shivaji College

University of Delhi, New Delhi, India

07/2007 – 06/2010

ACADEMIC APPOINTMENTS

Lecturer, NAS, Mahidol University, Thailand (1+ Year). **April 2025–Present**

Key Contributions:

- *Leader of laboratory in Sensor Intelligence, Data Modelling and Complex Systems*
- *Developed international collaborations across healthcare, sensing, and sustainability*
- *Teaching and mentoring PhD physics coursework and students*
- *Convener for Interfaces (International Talk Series) and English & Personality Development Club*

Adjunct Professor, Chitkara University, India (2+ Years). **April 2024–Present**

Key Contributions:

- *Research mentoring and collaboration*
- *Curriculum development and Joint research initiatives*

Assistant Professor, University of Delhi, India (9+ Years). **August 2015–March 2025**

Key Contributions:

- *Delivered 15+ undergraduate and postgraduate courses & supervised 50+ undergraduate research projects*
- *Developed international collaborations and funded projects from the British Council, UK*
- *Established a research lab on nanomaterials for undergraduate students (Leader)*
- *Member of the Skill Enhancement Course Committee and Value-Added Course Committee of DU*
- *Convener Research Cell of BNC, DU (from 2021-2024) & Mental Health Club of BNC, DU (from 2024-2025)*
- *Chair International Conference Series: “Vasudhev Kutumbkam” (2021, 2022, 2023, 2024 Edition)*
- *Editor Student Magazine of Research Cell and Mental Health Club, BNC (2021-2023)*
- *Convener, Institute innovation cell under Ministry of Education, Government of India, BNC, DU (2023-2025)*

Guest Professor, Department of Anthropology, University of Delhi. **September 2024–Present**

Key Contributions:

- *Delivered courses on Forensic Physics to the Master's in Forensics course and Certificate course*

Guest High School Teacher, Preet Public School, New Delhi. **April 2022–March 2023**

- *Delivered Standard XI Physics Course and set up Physics Laboratory*

Subject Matter Physics, Chegg.com, Online. **April 2016–March 2023**

- *Designing and solving Physics problems*

PROFESSIONAL AND EDITORIAL APPOINTMENTS

Editor, Scientific Reports, Nature Portfolio (2+ Years). **April 2024–Present**

- *Editorial assessment of multidisciplinary manuscripts and oversight of peer-review processes*

Associate Editor, ECS Sensor Plus, IOP Portfolio (2+ Years). **April 2023–August 2025**

- *Editorial assessment of multidisciplinary manuscripts and oversight of peer-review processes*

Review Editor, Frontiers in Nanotechnology, Frontiers **April 2023–February 2025**

- *Editorial assessment of multidisciplinary manuscripts and oversight of peer-review processes*

Guest Editor and Reviewer of Different Journals **April 2020–Present**

Special Issue: *IOP Nanotechnology, Results in Engineering, Materials Today proceedings, ECS JSS, Journal of Electrochemical Society, Biomass Conversion and Biorefinery, Environmental Science and Pollution Research*

Reviewer (Key Journals): *Biosensors & Bioelectronics, Chemical Society Reviews, Advanced Materials, Nano-Micro Letters, Materials Today Journals, Nature Nanotechnology, ACS Nano, Applied Physics Letters, Chemical Engineering Journal, IEEE Sensors, Journal of Materials Chemistry A-B, Sensors and Actuators. B, Chemical, Nanoscale, Sensors & diagnostics, Sensor & Biosensing Research, Sustainable materials and technologies, The European Physical Journal B, The Journal of Physics and Chemistry of Solids, National*

Academy Science Letters, Nanotechnology, Microchemical Journal, Journal of the American Chemical Society, iScience, Sensors, International Journal of Hydrogen Energy, Expert review of anti-infective therapy, Critical reviews in analytical chemistry, Applied Research, Analytical Chemistry Letters, Journal of Optics.

Resource Person and Expert, National Testing Agency, India. April 2023–March 2025

Key Contributions:

- *Design examination papers and review designed papers for the National level tests in Physics (India).*

Observer Officer, Indira Gandhi National Open University, India. April 2019–March 2025

Key Contributions:

- *Exam controller and exam centre head, special officer to conduct fair exams*

Research Fellow, DST, Government of India. August 2012–March 2015

Key Contributions:

- *Completed project titled conducting polymers-based gas sensors: SR/S2/CMP-0099/2010*

Founder, Sensor Intelligence Network (SIN), worldwide. April 2025–Present

Key Contributions:

- *Framing the concept of Sensor Intelligence and supporting young scientists through a collaborative network*

Founder, Happy Mentals, Mental Health Club. August 2021–Present

Key Contributions

- *Promoting mental health awareness*

EDUCATION LEADERSHIP & CAPACITY BUILDING

Courses Developed

- *Developing business acumen in science students*
- *Biosensor for detecting antimicrobial resistance*
- *Physics at Nanoscale*
- *Physics of Life-Matter Interfacial Dynamics*
- *Data Capturing Protocols for Experimentalists*
- *Data Modelling: ML and Non-ML approaches*
- *Breathomics*
- *Sensor Intelligence*

Academic Capacity Building

- *PhD Supervision: 02*
- *Master's and Undergraduate Supervision: 50+*
- *Resource Person at Faculty Development Programs: 10+*
- *Resource Person at Educational Camps and certificates for Students: 10+*
- *Plenary, Keynote, and Invited Speaker at conferences and workshops: 50+*
- *Diversity, Inclusion, and Mental Health projects resource person: 10+*

RESEARCH IMPACT, FUNDING & TRANSLATIONAL ECOSYSTEM

Research Metrics

- *152+ research and review articles with 75.2% as first/corresponding/both author with 85+ publications in Q1 journals, including Top 1% and Top 10% journals as per Scimago: Appendix: List of Publications)*
- *9+ conference proceedings and 8+ book chapters*
- *8+ Books, including 4 authored (Pearson) and 4 edited (Springer and Bentham Science)*

Research Ecosystem

International Projects and Fundings:

- *Decoding Antimicrobial abuse in wastewater through biosensors*

2023-2025

Role: Principal Investigator

Grant: Going Global Partnerships Academia-Industrial Collaborative Grant (44000 GBP)

Funder: British Council, UK

Partners: Ulster University (UK) and NanolatticeX (Industry)

- **Enhancing commercial acumen and organisational capability in business (ECOBUSS).** **2021-2023**
Role: Principal Investigator
Grant: Going Global Partnerships Exploratory Grant (ID: 877799913) (15000 GBP)
Funder: British Council, UK
Partners: University of Central Lancashire, UK

Institutional Fundings:

- **Mahidol President Fund for PhD Scholar** **2023-2025**
Role: Principal Advisor
Funder: Mahidol University

- **Mahidol Research Cluster Grant Program** **2023-2025**
Grant: Integrative Collaboration to Advance Melioidosis Research
Role: Team Member
Funder: Mahidol University

- **Institutional Projects-10 for undergraduates (Global Partnerships)** **2019-2023**
Role: Principal Investigator
Funder: Research Cell, BNC, DU

Projects and Partners:

- To study awareness about Immunity (with Weill Medical College, USA)
- Role of Social media during COVID-19 pandemic: COVAXIN Vaccine (with Gadjah Mada University, Indonesia)
- Impact of COVID-19 pandemic on the mental well-being of Youngsters (with SVRM, AP, India)
- Impact of COVID-19 pandemic on Socio-economic conditions in Indian society (with CBGA India)
- Impact of COVID-19 pandemic on human behaviour (with University of Hyderabad, India)
- Impact of COVID-19 pandemic on research and development of nanotechnology-based clean energy devices (with TIFR Mumbai, India)
- Analysis of online education system success rate amid COVID-19 pandemic (with IGNOU, India)
- Impact of COVID-19 and open book examination on social and mental well-being of final year students (with GGSIPU, Delhi, India)
- Assessing the challenges faced by health care workers during COVID-19 (with Max Planck's Institute, Germany)
- Role of print media in spreading awareness during COVID-19 pandemic (with University of Mumbai, India)

Social Fundings:

- **The Mental Health Project LGBTQIA+ Community in Science** **2021-2024**
Role: Principal Investigator
Funder: Happy Mentals Club and Elsevier Foundation

Partnership Ecosystem

Key Academic Collaborators: Xidian University (China), UTEP (USA), Florida Polytechnic University (USA), University of North Carolina and North Carolina State University (USA), Cornell University (USA), Ulster University (UK), University of Glasgow (UK), University of South Africa (South Africa), Yamagata University (Japan), Dublin City University (Ireland), University of Naples Federico II (Italy), SDU (Denmark), University of Delhi (India), University of Hyderabad (India), Shoolini University (India), Universiti Kebangsaan Malaysia, Jawaharlal Nehru University (India), Inha University (South Korea), Dongguk University (South Korea)

Key Clinical Collaborators: Safdarjung Hospital, Indian Council of Medical Research, Shantou University Medical College

Key Industrial Collaborators: NanolatticeX, SenseX

Key Network Collaborators: GFSN (UK), FAN (Futures AMR Network) (Europe), Sensor Intelligence Network

Important Media Coverage: Dainik Jagran (National Newspaper, India) – 10 times, AbhiTak India Magazine

Full List of Publications: (Impact Factor-wise)

1. H. Sable, V. Kumar, V. Singh, S. Rustagi, S. Chahal, **V. Chaudhary**, [Strategically engineering advanced nanomaterials for heavy-metal remediation from wastewater](#), *Coordination Chemistry Reviews*, 518, 2024, 216079. (IF: 25.6 Q1 - **Corresponding Author**)
2. B. Kaur, P. Rana, P. Singh, A. Singh, **V. Chaudhary**, S. Kaya, Q.V. Le, V. Nguyen, P. Raizada, [Unveiling new horizons of progress on manipulating the structure and characterization of phosphate-modified polymer for selective uranium adsorption](#), *Coordination Chemistry Reviews*, 518, 2024, 216057. (IF: 25.6 Q1, **Co-Author**)
3. S. Sambyal, A. Sudhaik, S. Sonu, P. Raizada, **V. Chaudhary**, V. Nguyen, A. A. P. Khan, C. M. Hussain, P. Singh, [Recent updates on cadmium indium sulfide \(CdIn₂S₄ or CIS\) photo-catalyst: Synthesis, enhancement strategies and applications](#), *Coordination Chemistry Reviews*, 535, 2025, 216653, (IF: 25.6 Q1, **Co-Author**).
4. **V. Chaudhary**, A. Gautam, P. Silotia, S. Malik, R.O. Hansen, M. Khalid, A. Khosla, A. Kaushik, Y. Mishra, [Internet-of-Nano-things \(IoNT\) driven intelligent face masks to combat airborne health hazards](#), *Material Today*, 60, 201-226 (2022). (IF: 24.1 Q1, ISSN: 1369-7021). (**First Author and corresponding Author**)
5. Abhimanyu, K. Singh, H. Sable, S. Sonu, V. Kumar, P. Singh, A. Kaushik, P. Thakur, **V. Chaudhary**, [Exploring complex silver sulfide nano-systems as efficient photocatalysts for wastewater remediation and clean energy production: Toward sustainable and circular economy](#), *Materials Today*, 90, 681-705 (2025). (IF: 24.1 Q1, **Corresponding Author**)
6. **V. Chaudhary**, S. Saichaemchan, P. Bhadola, A. Kaushik, [Nanostructured interfaces integrated with unsupervised intelligence to mitigate global polycrystalline complexities](#), *Advances in Colloid and Interface Science*, 356, 103970 (2026). (IF: 23.7 Q1, ISSN: 0001-8686, **Corresponding author**)
7. **V. Chaudhary**, S. Sonu, P. Raizada, A. Kaushik, [Bacteriogenic metallic and semiconducting nano-system as a potential sustainable solution for one health complexities](#), *Advances in Colloid and Interface Science*, 346, 103648 (2025). (IF: 23.7 Q1, **Corresponding author**)
8. K. Singh, Abhimanyu, Sonu, **V. Chaudhary**, P. Raizada, S. Rustagi, P. Singh, P. Thakur, V. Kumar, A. Kaushik, [Defect and Heterostructure engineering assisted S-scheme Nb₂O₅ nanosystems-based solutions for environmental pollution and energy conversion](#), *Advances in Colloid and Interface Science*, 332, 2024, 103273. (IF: 23.7 Q1, **Corresponding author**)
9. **V. Chaudhary**, N. Ashraf, M. Khalid, R. Walvekar, A. Kaushik, Y. Yan, Y. K. Mishra, [Emergence of MXene-polymer Hybrid Nanocomposites as High-Performance Next-generation Chemiresistors for Efficient Air Quality Monitoring](#), *Advanced Functional Materials*, 32, 2112913, 2022. (IF: 19.9 Q1, **First Author and Corresponding Author**)
10. N. Bhalla, M. Rabiey, P. Bendale, K. Lawther, J. Spencer, A. Longo, L. Lucky, **V. Chaudhary**, P. McCormack, S. Jana, P. SM Dunlop, L. Oyama, [From the lens of early-career researchers: bridging science, technology, arts, and humanities to tackle antimicrobial resistance](#), *Nature Communications* 17, 112 (2026). (IF: 18.1 Q1, **Co-Author**)
11. **V. Chaudhary**, Sonu, B.A. Taha, P. Raizada, S. Rustagi, S. Chahal, P. Singh, A. Khosla, V-H. Nguyen, [Borophene-based nanomaterials: Promising candidates for next-generation gas/vapor chemiresistors](#), *Journal of Materials Science & Technology*, 218, 236-262 (2025). (IF: 16.8 Q1, **First author & corresponding author**)
12. **V. Chaudhary**, E. Mostafavi, A. Kaushik, [De-coding Ag as an efficient antimicrobial nano-system for controlling cellular/biological functions](#), *Matter*, 5(7), 1995-1998 (2022). (IF: 15.7 Q1, **First Author and corresponding Author**)
13. **V. Chaudhary**, C. Muniraju, H. M. Albert, P. Bhadola, [A Simulation-Based Study of a Black Phosphorus-Based Complex Multilayer SPR-Based Optical Sensor for Wastewater Monitoring](#), *Electron*, 4(1); e70023 (2026) (IF: 15.2 Q1, **First- Corresponding Author**)
14. **V. Chaudhary**, H. Sable, Sonu, L. Phor, R. Umapathi, P. Bhadola, [Nanophotonic Biosensors and the Paradigm of Trans-Scale Autonomous Bio-Integrated Diagnostics for Addressing Health Complexities](#), *Electron*, e70043 (2026) (IF: 15.2 Q1, **First and Corresponding Author**)
15. **V. Chaudhary**, H. Sable, M. Kumar, C. Prakash, S. Sonu, [Dynamic landscape of chemiresistive breathomic nanosensors based on fifth-generation chips for complex disease diagnosis and healthcare monitoring](#), *Applied Physics Reviews* 2025; 12 (4): 041311. (5YIF: 15.2 Q1, **First Author and corresponding Author**)
16. P. Bhadola, H. Sable, P. Kumari, S. Kadian, R. Narayan, **V. Chaudhary**, [Sensor-driven modeling of complex systems using physics-informed supervised machine intelligence](#), *Applied Physics Reviews* 2026; 13, 021328, 2026. (5YIF: 15.2 Q1, **Corresp. Author**)

17. Kadian, S., Shukla, S., Yadav, A.K., Arya, B., Sethi, S., **V. Chaudhary**, and Narayan, R. (2025), [Recent Advancements in Graphene Quantum Dot-Based Bioimaging and Drug Delivery Systems](#). *MedComm*, 6: e70320. (IF: 14.1 Q1, Co-Author)
18. A. Khosla, Sonu, H.Taimoor, K. Singh, Gaurav, R.Walvekar, Z. Zhao, A. Kaushik, M. Khalid, **V. Chaudhary**, [Emergence of MXene and MXene-polymer hybrid membranes as future-environmental remediation strategies](#), *Advanced Science*, 9 (36), 2203527 (2022). (IF: 14.1 Q1, **Corresponding Author**)
19. K. Poonia, V. Soni, P. Singh, **V. Chaudhary**, S. Thakur, V. Nguyen, Q.V. Le, P. Raizada, [Metal-free defects in 2D Nanocarbons toward photocatalytic processes: Revolutionizing solar-to-energy conversion](#), *Applied Energy*, 382, 125195 (2025). (IF: 12.2 Q1, **Co-Author**)
20. **V. Chaudhary**, V. Khanna, H.Taimoor, M. Khalid, Y. Mishra, A. Kaushik, [Towards Hospital-on-chip supported by 2D MXenes-based 5th generation intelligent biosensors](#), *Biosensors & Bioelectronics*, 220, 114847 (2023). (IF: 11.8 Q1, **First author and corresponding author**)
21. B.A. Taha, A.J. Addie, E.M. Abbas, B.H. Aubaidan, N.M. Ahmed, A.J. Haider, **V. Chaudhary**, N. Arsad, [Biophotonics and nanorobotics for biomedical imaging, biosensing, drug delivery, and therapy](#), *Journal of Photochemistry and Photobiology C: Photochemistry Reviews*, 60–61, 100678 (2024). (IF: 11.6 Q1, **Correspon. Author**)
22. S. Banitaba, S. Ebadi, P. Sailmi, A. Bagheri, A. Gupta, W. Arifeen, **V. Chaudhary**, Y.K. Mishra, A. Kaushik, E. Mostafavi, [Biopolymer-based Electrospun Fibers in Electrochemical Devices: Versatile Platform for Energy, Environment, and Health Monitoring](#), *Materials Horizons*, 9(12), 2914-2948 (2022). (IF: 11.4 Q1, **Co-Author**)
23. **V. Chaudhary**, S. Rustagi, A. Kaushik, [Bio-derived smart nanostructures for efficient biosensors](#), *Current Opinion in Green and Sustainable Chemistry*, 2023, 42, 100817. (IF: 11 Q1, **First and Corresponding Author**)
24. **V. Chaudhary**, B.A. Taha, Lucky, S. Rustagi, A. Khosla, P. Papakonstantinou, N. Bhalla, [Nose-on-Chip Nanobiosensors for Early Detection of Lung Cancer Breath Biomarkers](#), *ACS Sensors*, 9 (9), 4469-4494 (2024). (IF: 10.9 Q1, **First author and corresponding author**)
25. M. Abdah, H.A. Taimoor, M.W. Alam, M. Mehar, M.N. Mustafa, R. Walvekar, M.W. Alam, M. Khalid, R. Umapathi, **V. Chaudhary**, [Advancements in MXene-polymer composites for high-performance supercapacitor](#), *Journal of Energy Storage*, 63, 2023, 106942. (IF: 10.7 Q1, **Corresponding Author**)
26. S. Subramanyam, L. Phor, Suman, **V. Chaudhary**, V. Kaushik, P. Kumar, S. Chahal, [Progress in MXene synthesis approaches for energy systems: A comprehensive review](#), *Journal of Energy Storage*, 92, 2024, 112043. (IF: 10.7 Q1, **Co-author**)
27. H.T.A. Awan, M.A.A.M. Abdah, M. Mehar, R. Walvekar, V. Chaudhary, M. Khalid, A. Khosla, [MXene-polymer hybrid composites for advanced energy storage: Insights into supercapacitors and batteries](#), *Journal of Energy Storage*, 95, 2024, 112449. (IF: 10.7 Q1, **Co-author**)
28. V. Batra, I. Kaur, Sonu, D. Pathnania, **V. Chaudhary**, [Efficient dye degradation strategies using green synthesized ZnO-based nanoplatfroms: A review](#), *Applied Surface Science Advances*, 11, 100314 (2022). (IF: 10 Q1, **Corresponding Author**).
29. **V. Chaudhary**, H. Sable, P. Bhadola, [Harnessing complex light-matter interactions for point-of-care nano-optical biosensing](#), *Advances in Physics: X*, 11(1), 2026. (IF: 9.9 Q1, **First author and corresponding author**)
30. **V. Chaudhary**, P. Gaur, S. Rustagi, [Sensors, society, and sustainability](#). *Sustainable Materials and Technologies*, 40, 2024, e00952. (IF: 9.7 Q1; **First & corresp. author**)
31. K. Singh, V. Khanna, A. Rosenkranz, **V. Chaudhary**, Sonu, G. Singh, S. Rustagi, [Panorama of physico-mechanical engineering of Graphene-reinforced Copper composites for sustainable applications](#), *Materials Today Sustainability*, 100560 (2023). (IF: 9.4 Q1, **Corresponding Author**).
32. **V. Chaudhary**, R.K. Talreja, S. Rustagi, R. Walvekar, A. Gautam, [High-performance H2 sensor based on Polyaniline-WO3 nanocomposite for portable batteries and breathomics-diagnosis of irritable bowel syndrome](#), *International Journal of Hydrogen Energy*, 52, 1156-1163 (2024). (IF: 9.2 Q1, **First & Corresponding Author**)
33. **V. Chaudhary**, Lucky, H. Sable, N. Bhalla, [Interdisciplinary Approach to Monkeypox Prevention: Integrating Nanobiosensors, Nanovaccines, Artificial Intelligence, Visual Arts, and Social Sciences](#), *Small Structures* 6, 2400647 (2025). (IF: 8.9 Q1, **First and Corresponding Author**).
34. **V. Chaudhary**, H.A. Taimoor, P. Badola, M. Khalid, A. Khosla, [Progress in engineering interlayer space modulated MXenes to architect next-generation airborne pollutant sensors](#), *Sensors and Actuators B: Chemical*, 379, 133225 (2023). (IF: 8.3 Q1, **First author and corresponding author**).
35. B. Barsola, S. Saklani, D. Pathania, P. Kumari, Sonu, S. Rustagi, P. Singh, P. Raizada, T.S. Moon, A. Kaushik, **V. Chaudhary**, [Exploring bio-nanomaterials as antibiotic allies to combat antimicrobial resistance](#), *Biofabrication*, 16 (4), 042007 (2024). (IF: 8.2 Q1, **Corresponding author**)

36. M. Mathur, A. Verma, A. Singh, B.C. Yadav, V. Chaudhary, [CuMoO₄ nanorods-based acetone chemiresistor-enabled non-invasive breathomic-diagnosis of human diabetes and environmental monitoring](#), *Environmental Research*, 2023, 229, 115931. (IF: 8.2 Q1, Corresponding Author)
37. V. Chaudhary, Sonu, R. Chowdhury, P. Thukral, D. Pathania, S. Saklani, Lucky, S. Rustagi, A. Gautam, Y.K. Mishra, P. Singh, A. Kaushik, [Biogenic green metal nano systems as efficient anti-cancer agents](#), *Environmental Research*, 2023, 229, 115933. (IF: 8.2 Q1, First and Corresponding Author)
38. P. Bhadola, V. Chaudhary, K. Markandan, R.K. Talreja, S. Aggarwal, K. Nigam, M. Tahir, A. Kaushik, S. Rustagi, M. Khalid, [Analysing role of airborne particulate matter in abetting SARS-CoV-2 outbreak for scheming regional pandemic regulatory modalities](#), *Environmental Research*, 236, 116646 (2023). (IF: 8.2 Q1, Corr. Author)
39. A. Jha, B. Barsola, D. Pathania, P. Raizada, P. Thakur, P. Singh, S. Rustagi, A. Khosla, V. Chaudhary, [Nano-biogenic heavy metals adsorptive remediation for enhanced soil health and sustainable agricultural production](#), *Environmental Research*, 252 (3), 2024, 118926. (IF: 8.2 Q1, Corresponding Author)
40. Nikita, M. Sharma, P. Thakur, P. Gaur, G.M. Rani, S. Rustagi, R.K. Talreja, V. Chaudhary, [Cancer treatment and toxicity outlook of nanoparticles](#), *Environmental Research*, 237, 116870 (2023). (IF: 8.2 Q1, Corresponding Author)
41. A. Jha, D. Pathania, Sonu, B. Damathia, P. Raizada, S. Rustagi, P. Singh, G.M. Rani, V. Chaudhary, [Panorama of biogenic nano-fertilizers: A road to sustainable agriculture](#), *Environmental Research*, 2023, 116456. (IF: 8.2 Q1, Corresponding Author)
42. V. Soni, M. Malhotra, A. Singh, V. Chaudhary, P. Singh, T. Aahmad, S. Kaya, C. M. Hussain, P. Raizada, [Emerging bismuth stannate semiconductor and its photocatalytic applications in pollutant degradation via Z/S-scheme heterostructures](#), *Environmental Research*, 279 (2), 121670 (2025). (IF: 8.2 Q1, Co Author)
43. B.A. Taha, Y.A. Mashhadany, Q. Al-Jubouri, A.R.B.A. Rashid, Y. Luo, Z. Chen, S. Rustagi, V. Chaudhary, N. Arsad, [Next-generation nanophotonic-enabled biosensors for intelligent diagnosis of SARS-CoV-2 variants](#), *Science of The Total Environment*, 880, 2023, 163333. (IF: 8.0, Corresponding Author)
44. Sonu, G.M. Rani, D. Pathania, Abhimanyu, R. Umapathi, S. Rustagi, Y.S. Huh, V.K. Gupta, A. Kaushik, V. Chaudhary, [Agro-waste to sustainable energy: A green strategy of converting agricultural waste to nano-enabled energy applications](#), *Science of The Total Environment*, 875, 2023, 162667. (IF: 8.0, Corresponding Author)
45. Y. Seth, S. Dharaskar, V. Chaudhary, M. Khalid, R. Walvekar, [Prospects of titanium carbide-based MXene in heavy metal ion and radionuclide adsorption for wastewater remediation: A review](#), *Chemosphere*, 2022, 133563. (IF: 8.0, Co-author)
46. Soni V., Sudhaik A., Singh P., Thakur S., Ahamad T., Nguyen V., Thi L.P., Quang H.H.P., V. Chaudhary and Raizada P. [Visible-light-driven photodegradation of methylene blue and doxycycline hydrochloride by waste-based S-scheme heterojunction photocatalyst Bi₅O₇I/PCN/tea waste biochar](#). *Chemosphere*, 347, 140694, 2024. (IF: 8.0; Co-author)
47. V. Chaudhary, P. Bhadola, [Artificial Intelligence-Powered Nanosensor Platforms for Non-Invasive Breathomic Diagnostics](#). *Nanotechnology, Science and Applications*, 18, 611–641 (2025). (IF 7.6, Q2, First and Corresponding Author)
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